

Energy and exergy analysis of a hybrid active-passive PV cooling system with MgO-enhanced PCM, metallic microchannels, and TEGs

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ABSTRACT

Photovoltaic thermal (PVT) systems, which convert solar energy into both electricity and heat, face performance challenges due to rising temperatures. Phase change materials (PCMs) offer thermal storage benefits but suffer from low thermal conductivity. To address this, lauric acid (LA) was enhanced with magnesium oxide (MgO) nanoparticles, forming MgO-enhanced phase change material (MgO-PCM). A novel container with micro-channels and eight interconnected water tubes was designed to improve heat transfer and electrical efficiency. Waste heat from the PV module is absorbed by the MgO-PCM and transferred to the water tubes. A thermoelectric generator (TEG) further converts excess heat from the water outlet into electricity. System performance was tested at tilt angles of 15°, 30°, 45°, and 60°, using an optimized inlet water flow ($Re = 5500$). Results indicated that a 15° tilt offered the best performance. Compared to an uncooled PV system, the PVT system achieved a 29.08% increase in electrical efficiency, 70% thermal energy recovery, 82.49% overall efficiency, and 12.78% exergy efficiency.

1. Introduction

Rising worldwide energy demand and concern about global warming, energy conservation, and emission reduction had become global research priorities, and scientists had carried out multiple investigations. Kabir et al. [1] explored the technical problem affecting renewable energy research and their future prospect. The roadmap for the field of solar research was thoroughly discussed. Key summary was the solar energy system as a standalone solution could be feasible with high storage temperatures. Over the past decade, global solar photovoltaic (PV) capacity deployment has grown, demonstrating that PV systems were reducing carbon emissions and energy consumption. Bennouna et al. [2] found that decentralized solar energy production allow peak clipping by more than 12.6% of yearly maximum in Morocco for the year 2018. Photovoltaic (PV) systems turned sunlight into energy

at 20% efficiency; however, they absorbed over 50% of sunlight as heat [3]. Unfortunately, photovoltaic modules lost efficiency when exposed to high temperatures [4]. This thermal gain raised the PV cell operating temperature. According to Ref. [5], a 1 °C PV cell temperature increases reduced conversion efficiency by 0.08% and power production by 0.45%. Therefore, scientists have studied and developed several methods to regulate photovoltaic (PV) system temperature. Waste heat recovery systems can convert PCM waste heat into useful energy. Some solutions included thermoelectric generators. TEG generated power from PCM's excess heat. This combination can boost power generation and reduce solar cell use, lowering expenses. These eco-friendly, non-toxic solutions were inexpensive [6].

There were 8135 publications related to “photovoltaic cooling” based on the Scopus database from 1995 to 2024. The number of articles consistently increased, reaching its peak in the latest year with an

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increasing interest in the study of passive cooling. Water cooling had approximately twice the number of publications compared to air cooling. Additionally, phase change material (PCM) was commonly used to reduce the temperature of photovoltaic (PV) modules in passive cooling systems. Incorporating nanoparticles with high thermal conductivity can improve the heat dissipation capability of a PCM. This approach allowed for a comprehensive view and scope of research, making it possible to identify new research gaps.

Implementing phase change material (PCM) could improve performance by as much as 21.2% [7]. Numerous researchers had expressed interest in the potential of PCM to aid in the cooling of PV modules. Arıcı et al. [8] found that the operating temperature of PV panel could be reduced up to 10.26 °C with an efficiency improvement up to 3.73%. Hindi et al. [9] designed and experimented on a performance of the PV system by using paraffin as a PCM placed into an aluminum container with internal and exterior longitudinal aluminium fin. The result was an increase in the average maximum power output up to 23.1%. PV modules enabled heat transfer to PCMs without necessitating an external working fluid. A container secured the PCMs behind the Tedlar surface. Velmurugan et al. [10] did a review of various types PCM and suggested a cascaded PCM for PV under tropical climate conditions. The literature reported many successful attempts to improve the performance of PV-PCM systems. For example, Khanna et al. studied finned PV-PCM systems, observing that increasing the cavity thickness extended the panel cooling period [11]. Wongwattanasatian et al. [12] also did an experiment with PCM (palm wax) inside three different containers (grooved, tubed, and finned) to control the temperature of PVs that were exposed to less than 1000 W/m² of sunlight every day. The finned PCM container performed best, lowering panel temperature by 6.1 °C with an increase in cell efficiency of 5.3%. Homlakorn et al. [13] had studied eutectic PCMs made from fatty acids and filling a mesh-finned PV module system PCM container. Electrical output rose 5.04 percent. Bestas et al. [14] added different amounts of nanoparticles (Al₂O₃) to PCM (RT35) to help lower the surface temperature of PV panels and improve their performance. The investigation analyzed five systems: a reference PV panel (PV), a PV panel utilizing PCM without nanoparticles (PVPCM-0), and PV panels with PCM incorporating varying concentrations of nanoparticles (PVPCM-0.05, PVPCM-0.1, and PVPCM-0.15). Among the five different systems, the PV panel with 0.15% w/v nanoparticles (designated as PVPCM-0.15) demonstrated the most effective cooling capability. Furthermore, the PVPCM-0.15 system demonstrated superior performance, achieving a 19.49% enhancement in panel power output, along with averaged energy and exergy efficiency values of 10.44% and 7.29%, respectively, for PVPCM-0.15. Bhutto et al. [15] examined the use of silver (Ag) and graphene (Gr) hybrid nanoparticles (NPs) as additives to enhance the thermophysical properties of lauric acid (LA) phase change material (PCM). The nanocomposite LAG-3.5 consisted of LA with 3.5% hybrid Ag and Gr NPs, intended for use in photovoltaic (PVT) and thermoelectric generator (TEG) applications. The findings indicated that the NePCM-integrated PVT system lowered the PV surface temperature to 39 °C, whereas the PV system reached 63 °C. The nanocomposite LAG-3.5 showed a higher electrical efficiency of 9.42% when used in a nano-enhanced PCM-based photovoltaic thermal system, while the photovoltaic system alone had an efficiency of 6.68%. Lastly, many PCMs had limited thermal conductivity. Thermal additives improved heat transport and cooling, although they lowered latent heat of fusion and increased weight [16].

A variety of PVT configurations were examined utilizing literature. This analysis can help create a new PVT system by selecting the best configuration for each application. Work by Gaur et al. [17] examined a forced-air flat-plate monocrystalline glazed PVT system. BiSPVT: This semitransparent system was fitted into a building. This system's semitransparency and integration with a building made it a BiSPVT system. The air duct-equipped BiSPVT system had an average daily electrical efficiency of 13.11% and a room temperature of 10.1 °C. Compared to systems without ducts, air-duct systems generated 9.8% more electrical

and 30% more thermal energy. The micro-channel PVT module developed by Zhou et al. [18] functioned as an evaporator in a direct expansion heat pump system. It had 13% electrical efficiency and 56.6% thermal efficiency in experiments. Simulations of Yao et al. [19] indicated that high-altitude heating can be achieved by combining a PVT collector with an air-source heat exchanger in direct expansion. The thermal efficiency of the system was 55.76%, while the electrical efficiency was 17.77%. A hybrid system (PV/T-PCM-TEG) using a 40 W PV panel was the subject of a study in 2023. To enhance the efficiency of the PV/T-PCM-TEG system, three distinct scenarios were developed: Case 1 utilized RT55 paraffin wax as the phase change material, Case 2 incorporated Al₂O₃ nanoparticles into the phase change material, and Case 3 featured copper fins surrounding the copper pipes. The efficiency of PV/T-PCM-TEG was raised to 32.8%, 33.9%, and 35.2% for Case 1, Case 2, and Case 3, respectively [20]. El-Nagar et al. [21] looked at both active and passive cooling methods in photovoltaic (PV) systems to improve the power output and thermal efficiency of building-integrated photovoltaic (BIPV) modules. A number of tests were done using five different modules: one standard module, a second one that was a BIPV/T system with cooling water, and three BIPV/T-PCM systems that had different setups of PCM cylindrical capsules placed on the back of the photovoltaic panels, all surrounded by cooling water channels. The BIPV/T module with a 40% volume ratio of PCM capsules to cooling water channels reduced temperature by 29.6 °C (37.4%) and had the maximum overall efficiency of 73.02%. Kang et al. designed and analyzed a BIPV combined with PCM and TEG (BIPV-TEG-PCM) through computational fluid dynamics using Ansys Fluent R1 and PiAnO [22].

Many studies used thermoelectric generators (TEG) to boost electrical output, whether in a PV or PVT system. There were 280 relevant publications from the last ten years in the Scopus database. Some studies combined solar energy and thermoelectric power in the PVT system to increase energy conservation and efficiency. Almarashi et al. [23] studied the effect on TEG performance by adjusting nanoparticle concentration, Hartmann number, and inlet velocity. Azizi et al. [24] designed a PVT system to include an Fe₃O₄/SiO₂ hybrid nanofluid mixture inside a cooling tube equipped with anchor-shaped fins. Additional glass coating by SiO₂ nanoparticle was implemented to reduce the impact of dust deposit which improved the temperature uniformity across the panel. Babu and Ponnambalam designed a non-concentrated flat plate PV-TE with additional 5% energy and 6% efficiency. Various applications include electricity generation in extreme environments, waste heat recovery in the transport industry, micro-generation and solar thermoelectric generators [25]. It can generate electricity using the Seebeck effect or chill the system using the Peltier effect as a solid-state direct energy converter. Matheswaran et al. [26] found that hybrid systems can improve conversion efficiency by integrating PV and TEG modules. These modules converted solar energy into electricity. Lekbir et al. [27] developed a PV/T-TEG hybrid system using nanoparticles (nanofluids) to improve hybrid PV and TEG modules. Research showed that CPV/T-TEG systems with nanoparticles produced 10.01%, 47.69%, and 49.49% more power than nanoparticles, CPV, and CPV/TEG systems. Wen et al. [28] also statistically investigated solar irradiation concentration using a linear Fresnel-concentrated photovoltaic/thermal-TEG (LFPV/T-TEG) system with or without a gravity-driven heat pipe (GHP) to capture waste heat. A study indicated that the GHP increased LFPV/T-TEG efficiency from 70.4% to 73.6% under 800 W/m² of light for 7 h at 18 °C. Additionally, Naderi et al. [29] investigated the possibility of PV-TEG with phase change materials (PCM) to improve performance and generate power. Compared to a solo PV, the PV temperature dropped from 74.43 °C to 53.22 °C, and its efficiency and electrical output rose by 100% and 1.38%. PVT/Water and PVT/TEG/nanofluid were studied in 2023. The PVT/Water and PVT/TEG/nanofluid setups dropped 25.1% and 41.2% in temperature and increased electrical efficiency by 5.8% and 8.5% over the reference PV panel [30]. Khanalizadeh et al. [31] introduced an innovative configuration of a photovoltaic/thermal-thermoelectric generator

(PV/T-TEG) integrated system, which employed water and air cooling methods to improve the efficiency of PV modules. The system utilized TEGs positioned between the water heat exchanger and the air heat sink, rather than placing them between the PV module and the cooling fluid. The findings demonstrated that the proposed system led to a 23% reduction in PV surface temperature, an increase in electrical output power exceeding 15%, and an enhancement in PV electrical efficiency of approximately 3%. Rad et al. [32] conducted a study to determine the potential impact of combining PCM and TEG with PVT units, as well as adding PCM and nanofluids to concentrated PVT (CPVT) units, on their performance. The findings revealed that the use of phase change materials, porous media, and concentrator plates increased energy generation, which might offset their higher starting costs and result in an energy cost of roughly \$0.135/kWh. Nonetheless, the use of TEG and nanofluids was proven to be economically unsustainable, resulting in a 15–35% increase in energy costs. The principal output achieved electrical, thermal, and exergy efficiencies of around 15.5%, 37.4%, and 16.7%, respectively, as well as a reduction in PV module temperature of more than 20%.

Several PVT cooling methods were studied, but the PVT, which used a microchannel PCM container and TEG, had not. TEG used the Seebeck phenomenon to directly convert thermal energy into electrical energy. The temperature differential between the connections also affected TEG efficiency [33]. Behind the PV module was the micro-channel PCM container with eight water tubes. The micro-channel container filled with MgO nanoparticles mixed with lauric acid absorbed waste heat from the PV module. Each water tube then cooled the PCM by transferring surplus heat into water. Waste heat recovery technologies such as thermoelectric generators can convert the PCM's waste heat into electricity.

Despite extensive research on PV modules with TEG, a thorough review of the literature revealed a knowledge gap in the application of PVT systems and the incorporation of TEG. The PVT system used the MgO-nanoparticle mixture of PCM (MgO-enhanced phase change material, MgO-PCM) and metallic micro-channel containers, along with TEG integration, to generate additional electricity and fully utilize the heat generated by the PV module. Furthermore, research on TEGs typically placed them beneath solar panels. In this study, TEGs were placed at the water tube outlet to generate additional power, thereby enhancing the system's overall efficiency. This container's distinguishing feature was its entirely filled micro-channel construction, which enabled MgO-PCM filling and the insertion of eight water tubes. The components inside this micro-channel PCM container, including MgO-PCM and a micro-channel structure, provided better heat transfer, thus lowering the surface temperature of the PV. For this investigation, a serpentine flow water tube configuration with a dimensionless flow type of 5500 was used because it gave the best overall performance. The main goals of the current study were to increase power generation and the overall efficiency of the PVT system. Additionally, this study aims to investigate the effects of varying inclinations (angles). Our study used different angles of tilt (15°, 30°, 45°, and 60°) and tested them with a PV module that had a micro-channel PCM container attached to the back of the panel to find the best configuration. The sum of electrical efficiency and thermal efficiency determines the optimal angle designed. It is anticipated that the TEG can extract heat from the water within the water tube outlet, thereby enhancing the temperature reduction of the PVT system and ultimately increasing the efficacy of the module.

2. Materials and methods

2.1. Inclinations

The main goals of this investigation were to determine the maximum power generated by varying inclinations and to increase the electrical efficiency of the PVT system by using thermoelectric generators (TEG). The TEG absorbed excess heat from the container, generating secondary

electricity. Therefore, this study tested a PV module at different levels of inclination, specifically 15°, 30°, 45°, and 60°. Each experiment for each tilt angle was conducted one day at a time and repeated for more than three days. Then three days that had similar solar irradiation values were selected to calculate the averaged results. Fig. 1 illustrated a schematic representation of the PVT, including the PCM container, the micro-channel PCM container, and a sectional view of the entire setup.

2.2. Materials

The current study combined lauric acid (LA), a PCM, with MgO, a metal oxide nanoparticle, to form a MgO-enhanced phase change material (MgO-PCM). Lauric acid was an organic fatty acid available in powder or flaked form. It had a density of 880 kg/m³, a melting point of 42.26 °C, a heat of fusion of 210 J/g, and a thermal conductivity of 0.094 W/m K. MgO particles were a type of nanoparticle commonly referred to as magnesium oxide. Magnesium oxide was a naturally occurring white powder with a particle size of 125 µm. It had a melting point of 2800 °C, a latent heat of fusion of 1775 J/g, and a thermal conductivity of 45 W/m K. The previous study [34] found that mixing lauric acid (LA) with three types of metal nanoparticles, Al₂O₃, MgO, and CuO, created 15 nanoparticle-enhanced PCMs in the following amounts: The subsequent proportions were as follows: 99:1, 98:2, 94:6, 92:8, and 90:10. The combination of PCM and nanoparticles led to an enhancement in thermal conductivity, while there was a reduction in latent heat. After thorough analysis of the balance point between thermal conductivity and the latent heat of fusion, the optimum nanoparticle-enhanced PCM was identified with a 94:6 ratio of LA:MgO. This study selected lauric acid (3.44 USD/kg) because it is 1.1–145.9 times cheaper than other commercial phase change materials (PCMs) and has a lower density and acceptable melting temperatures. Furthermore, in terms of cost, MgO nanoparticles (2.19 USD/kg) are 18.5 to 222.2 times cheaper than other commercially available nanoparticles. The comparison of material costs is indicated in Table 1. The thermal conductivity of PCM was considerably enhanced by this composition, with an estimated 65.96% increase. Consequently, this 94:6 ratio of MgO-PCM was selected for further investigation. It had a melting point of 41.28 °C, a latent heat of 167 J/g, and a thermal conductivity of 0.156 W/m K.

2.3. Water tube configurations and PCM-container

In our 2024 study, dimensionless levels (Reynolds number, Re) were used to represent the water flow rates. Seven distinct levels of dimensionless Reynolds number (Re), specifically 1100, 2200, 3300, 4400, 5500, 6600, and 7700, were examined using a photovoltaic (PV) module equipped with a micro-channel phase change material (PCM) container positioned at the rear of the PV panel. In order to facilitate the cooling of the PV module, a micro-channel PCM container was installed at the rear of the module. The rectangular stainless steel container had dimensions of 580 mm × 630 mm × 100 mm. The micro-channel container was filled with MgO-PCM in a grid of small square channels, each measuring 12.7 mm × 12.7 mm. There were eight water tubes, each having a diameter of 19.1 mm. These tubes were inserted between the micro-channels, spaced 80 mm apart. This investigation utilized three distinct configurations: a U-tube, a half-serpentine flow, and a serpentine flow. The results of the study revealed that the cases with the serpentine flow design and the water inlets with a Reynolds number (Re) of 5500 provided the maximum overall efficiency at 80.63%. Thus, a serpentine flow configuration with a Reynolds number (Re) of 5500 was selected for this study.

2.4. Experimental setup

The system used was the PVT-PCM system, where PVT was active cooling and PCM was passive cooling; thus, it was referred to as a hybrid

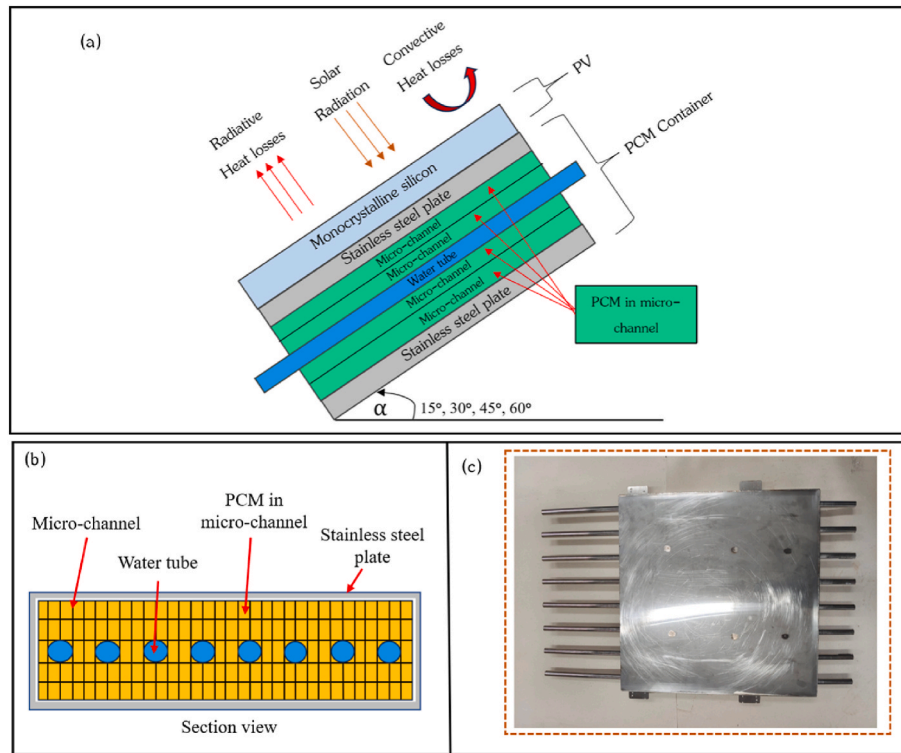


Fig. 1. (a) Schematic of the PVT with PCM container (b) Section view (c) micro-channel PCM Container.

Table 1
Comparison of material costs.

Materials	Costs (USD/kg)
Lauric acid	3.44
Stearic acid	3.86
Paraffin wax	5.79
Hydrocinnamic	502.2
Chloroacetic acid	89.04
Glauber's salt	114.6
Graphene	487.28
TiO ₂	94.75
SiC	175.96
CuO	31.19
Al ₂ O ₃	13.1
MgO	2.19

active and passive PV cooling. The study was done in outside environments to investigate how efficient a reference PV module was compared to one that has MgO-PCM inside a micro-channel PCM container. The system employed monocrystalline PV modules with a power output of 100 V, the module characteristics of which were listed in Table 2. The instruments employed to quantify temperatures were K-type

Table 2
PV module characteristics.

Module	Range
Maximum Power (P_{max})	100W
Maximum Power Current (I_{mp})	5.42A
Maximum Power Voltage (V_{mp})	18.22V
Short Circuit Current (I_{sc})	5.85A
Open Circuit Voltage (V_{oc})	21.62V
Production Tolerance	$0 \pm 3\%$
Dimension (L x W x D)	780 x 680 x 30 mm
Temperature coefficient of P_{max}	$-0.38\%/^{\circ}\text{C}$
Module efficiency	$>18\%$
NOCT (Nominal Operating Cell Temperature)	$45 \pm 2^{\circ}\text{C}$

thermocouples. The maximal voltage (V_{max}) and maximum current (I_{max}) were measured when the apparatus was connected to a resistive load (13.5Ω). Three thermoelectric generators (TEGs), each sized $40 \text{ mm} \times 40 \text{ mm} \times 3.4 \text{ mm}$, were set up to reduce the temperature of the water that exited the system, and the temperature difference between the hot and cold sides of the TEG produced an additional power output. The TEG was designed for a maximum temperature difference of 60°C and an operating temperature limit of 125°C . The pyranometer was used to measure the solar irradiance over the course of a day at Khon Kaen University, Thailand (Coordinates: 16.473057278994524, 102.82210165221844). Figs. 2 and 3 depicted the real experimental apparatus. The experiments were conducted one day at a time for each inclination angle and were repeated daily until the data showed the comparable solar irradiation values for the entire day. Only then would the data be included in the data set for the average results. Moreover, this research's PVT system included a very large water tank (3000 L) which is inside a building with an insulated roof. Additionally, after the water exits the system, it must travel a considerable distance (around 35 m) before returning to the water tank. Therefore, the water returning to the tank did not cause any significant changes to the water inlet in the PVT system. The experiments were done one day at a time for each tilt angle and were replicated over a period of several days. Then, three days with comparable solar irradiation values throughout the day were chosen for analysis. The objective was to guarantee that the experimental conditions were comparable, as illustrated in Table 3.

Fig. 4(a) depicted the process diagram of the experiment. Both PV modules, with and without cooling, were connected to a resistive load (13.5Ω). The current, voltage, and power were measured by multimeter. In addition, the surface temperatures of the PV module were measured by thermocouple. TEG equipment was also connected to a resistive load (1Ω). A multimeter and a thermocouple were used to measure the electrical parameters and the cold/hot temperatures of the TEG, respectively. Moreover, solar irradiation and ambient temperature were measured by pyranometer and ambient thermometer, respectively. All data was read and recorded in the data logger, after which the results were analyzed by the computer. Each configuration required a minimum

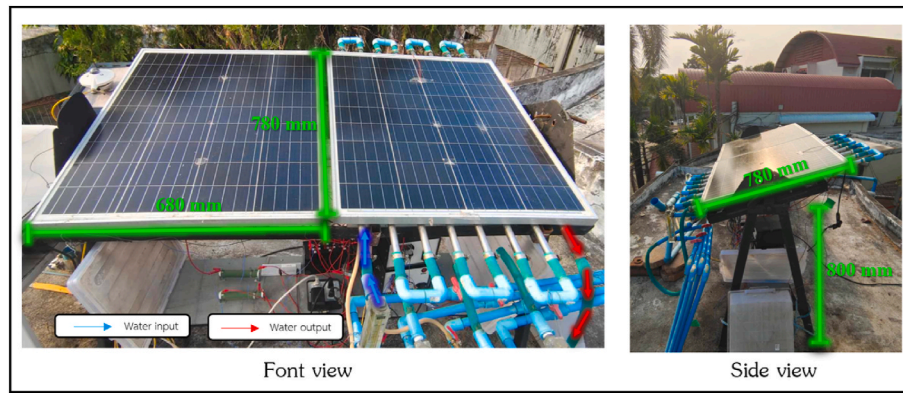


Fig. 2. PVT system with serpentine flow configuration.

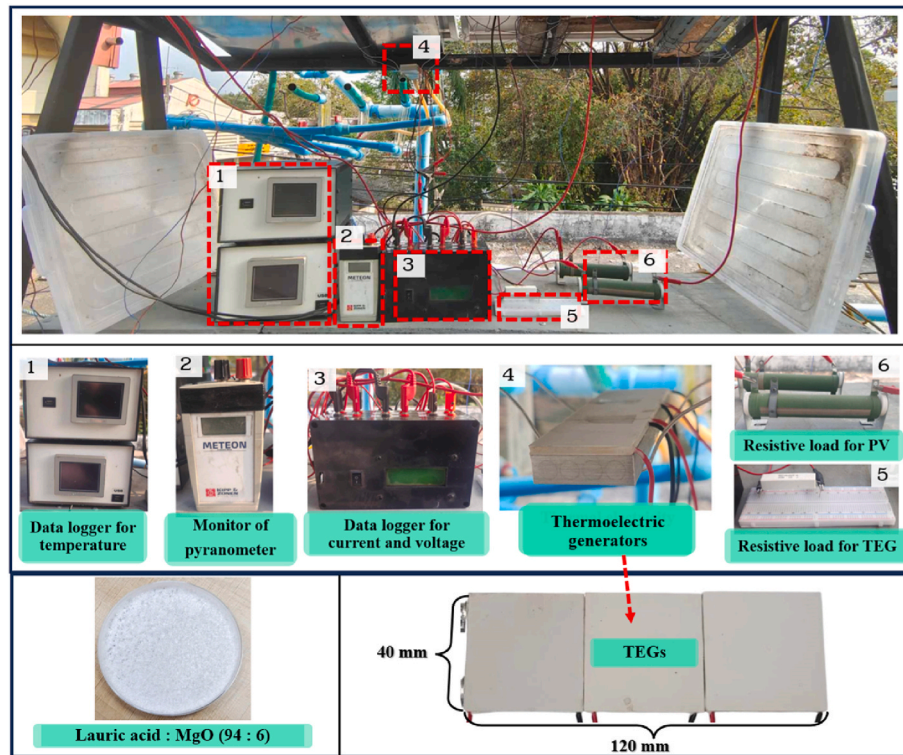


Fig. 3. Installation of experimental equipment.

of 3 days of comparable values. The average ambient temperature at the experimental site is 24.53°C . The solar irradiation during the day was measured, as well as the ambient temperature, as shown as a top orange line for light intensity and a bottom dark yellow line for ambient temperature in Fig. 6(a). The study's limitations encompass a variation in solar irradiation of approximately $\pm 0.65\%$ of the selected period of the experiment. Fig. 4(b) depicted a schematic diagram of the system. T1–T3 are temperature measurement points of the PV module without cooling (reference PV module), for which average data is calculated. T4–T6 are temperature measurement points for the PV module with the PCM container, for which average data is calculated. T7 is the PCM temperature measurement point, while T8–T10 are the temperature measurement locations for the PVT system's water inlet, TEG water inlet, and PVT system water outlet, respectively. T11 is the measurement point for ambient temperature. I1 and V1 are volt-ampere measuring points for the PV module without cooling (reference PV module). I2 and V2 are volt-ampere measurement points for the PV module with a PCM container, while I3 and V3 are volt-ampere measurement points for TEG

devices.

2.5. Uncertainty

Contributing to these errors were the experimenter, the measurement instruments, and the testing conditions. All experiments were carried out in three replications, and the average data were presented in the results. The devices' uncertainty and the statistical standard deviation (SD) were considered. These errors were acceptable and dependable for all tests. Moreover, error bars were added in every result's graphs. The measurement devices' uncertainty is indicated in Table 4.

2.6. Energy balance

The correlation between the temperature of the PV module and its electrical power profile was being determined by employing equations (1)–(4). These equations furnished us with the power, electrical efficiency (η_{el}), and thermal efficiency (η_{th}) of the modules under inquiry.

Table 3

Experimental dates.

Tilt angle	Date (November 2024)		
	1st Iteration	2nd Iteration	3rd Iteration
15°	1st $IR_{avg} = 780.5 \text{ W/m}^2$, $T_{amb-avg} = 24.4 \text{ }^\circ\text{C}$	2nd $IR_{avg} = 783.1 \text{ W/m}^2$, $T_{amb-avg} = 24.1 \text{ }^\circ\text{C}$	3rd $IR_{avg} = 772.8 \text{ W/m}^2$, $T_{amb-avg} = 24.5 \text{ }^\circ\text{C}$
30°	5th $IR_{avg} = 765.5 \text{ W/m}^2$, $T_{amb-avg} = 24.3 \text{ }^\circ\text{C}$	8th $IR_{avg} = 768.4 \text{ W/m}^2$, $T_{amb-avg} = 25.1 \text{ }^\circ\text{C}$	9th $IR_{avg} = 774.6 \text{ W/m}^2$, $T_{amb-avg} = 25.0 \text{ }^\circ\text{C}$
45°	10th $IR_{avg} = 778.1 \text{ W/m}^2$, $T_{amb-avg} = 24.4 \text{ }^\circ\text{C}$	15th $IR_{avg} = 779.6 \text{ W/m}^2$, $T_{amb-avg} = 24.4 \text{ }^\circ\text{C}$	16th $IR_{avg} = 775.3 \text{ W/m}^2$, $T_{amb-avg} = 24.6 \text{ }^\circ\text{C}$
60°	20th $IR_{avg} = 775.2 \text{ W/m}^2$, $T_{amb-avg} = 25.1 \text{ }^\circ\text{C}$	26th $IR_{avg} = 772.3 \text{ W/m}^2$, $T_{amb-avg} = 24.3 \text{ }^\circ\text{C}$	29th $IR_{avg} = 782.2 \text{ W/m}^2$, $T_{amb-avg} = 24.2 \text{ }^\circ\text{C}$

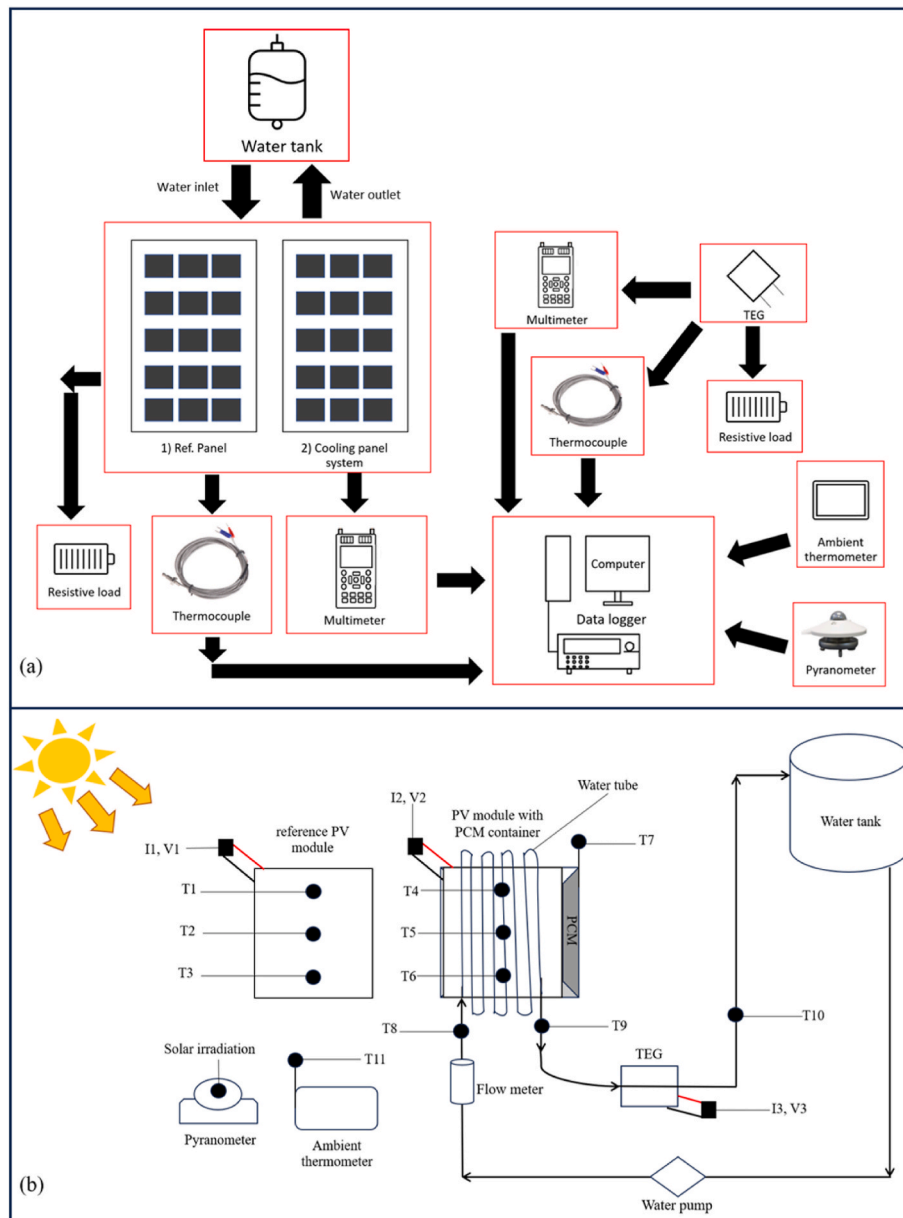
* IR_{avg} = The average solar irradiation, $T_{amb-avg}$ = The average ambient temperature.

This study oriented the PV modules towards the south and adjusted the PV tilt angle at 15°, 30°, 45°, and 60°. Additionally, the Reynolds number (Re), or dimensionless quantity, was transformed from the flow rate of the water input (volume flow rate) to establish a standard for future investigations. It used equations (5) and (6) to determine the Reynolds number and flow rate of the water inflow, respectively. Moreover, equations (7)–(11) determine the electrical efficiency of the TEG (η_{TEG}). Finally, equation (12) illustrated that the overall efficiency

Table 4

Uncertainty of the measurement devices.

Measurement device	Range	Uncertainty
K-type thermocouple	−270 °C ~1260 °C	±0.30%
Temperature data logger	-	±0.05%
Pyranometer	0 ~ 4000 W/m ²	±0.25%
Multimeter	200.0 mV ~600V, 200.0Ω ~20.00MΩ, 200.0 μA ~10.00A	±0.50%

**Fig. 4.** (a) Concept design of experimental RIG (b) Schematic diagram of the system.

($\eta_{overall}$) was the sum of the thermal efficiency, the electrical efficiency of the PV module, and the electrical efficiency of the TEG.

$$P = VI \quad (1)$$

$$\eta_{el} = \frac{P}{IR \times A} \times 100\% \quad (2)$$

$$\eta_{th} = \frac{\dot{Q}_U}{IR \times A} \times 100\% \quad (3)$$

$$\dot{Q}_U = (\dot{m}c_p)_w (T_{fo} - T_{fi}) \quad (4)$$

$$Q_{vf} = vA_t \quad (5)$$

$$Re = \frac{\rho v D}{\mu} \quad (6)$$

$$P_{TEG} = Q_{high} - Q_{low} = sa(T_{high} - T_{low})I - I^2R = \frac{sa^2(T_{high} - T_{low})^2}{4R} \quad (7)$$

$$Q_{high} = saT_{high}I + \frac{T_{high} - T_{low}}{R_t} - \frac{I^2R}{2} \quad (8)$$

$$Q_{low} = saT_{low}I + \frac{T_{high} - T_{low}}{R_t} + \frac{I^2R}{2} \quad (9)$$

$$I = \frac{sa(T_{high} - T_{low})}{2R} \quad (10)$$

$$\eta_{TEG} = \frac{P_{TEG}}{IR \times A} \times 100\% \quad (11)$$

$$\eta_{overall} = \eta_{el} + \eta_{th} + \eta_{TEG} \quad (12)$$

Where: P is the power output in watt (W), V is voltage in volts (V), I is current in amperes (A), IR is the solar irradiance in watt per square meter (W/m^2), A is area of the PV module in square meter (m^2), $\dot{Q}_U$ is useful heat rate from PVT module in kilojoule per second (kJ/s), $\dot{m}$ is mass flowrate in kilogram per second (kg/s), c_p is specific heat in

kilojoule per kilogram-degree Celsius ($kJ/kg^\circ C$), T_{fo} is temperature outlet in degree Celsius ($^\circ C$), and T_{fi} is the temperature inlet in degree Celsius ($^\circ C$), Q_{vf} is volume flowrate in cubic meter per second (m^3/s), v is velocity in meter per second (m/s), A_t is area of tube in square meter (m^2), ρ is density of the fluid in kilogram per cubic meter (kg/m^3), D is diameter of tube in meter (m), μ is dynamic viscosity in Pascal-second (Pa.s), P_{TEG} is the power output of TEG in watt (W), Q_{high} and Q_{low} are the heat flows at the hot side and the cold side of the TEG, respectively, in watts (W), sa is the Seebeck coefficient of the TEG in volts per degree Celsius ($V/^\circ C$), T_{high} and T_{low} are the temperature of the hot side and the cold side of the TEG, respectively, in degrees Celsius ($^\circ C$), R_t is the heat resistance of the TEG in degrees Celsius per watt ($^\circ C/W$), and R is the electrical resistance of the TEG in Ohm (Ω).

Fig. 5 depicted the conversion of solar energy at point 1 into electricity at point 2, as the photovoltaic surface loses energy. Following that, the phase change material within the micro-channel container gathered waste heat from the photovoltaic module and transferred it to water via the water tube. This procedure also resulted in an energy loss. The passage of high-temperature water through one side of the TEG created a temperature difference, which generated an electrical current. However, the TEG lost energy as well. Thermal energy was calculated by designating point A as the temperature of the water inflow and point C as the temperature of the water output.

2.7. Exergy analysis

The exergy analysis, which assessed the maximum work that can be extracted from the PVT system in relation to its environmental context, was conducted through the application of the second law of thermodynamics. The calculation of exergy loss (Ex_{loss}) and exergy efficiency (η_{Ex}) can be performed using [35]:

$$\eta_{Ex} = \frac{Ex_{out}}{Ex_{in}} \times 100\% \quad (13)$$

$$Ex_{in} = IR \times A \left(1 + \frac{1}{3} \left(\frac{T_{amb}}{T_{sun}} \right)^4 - \frac{4}{3} \left(\frac{T_{amb}}{T_{sun}} \right) \right) = Ex_{sun} \quad (14)$$

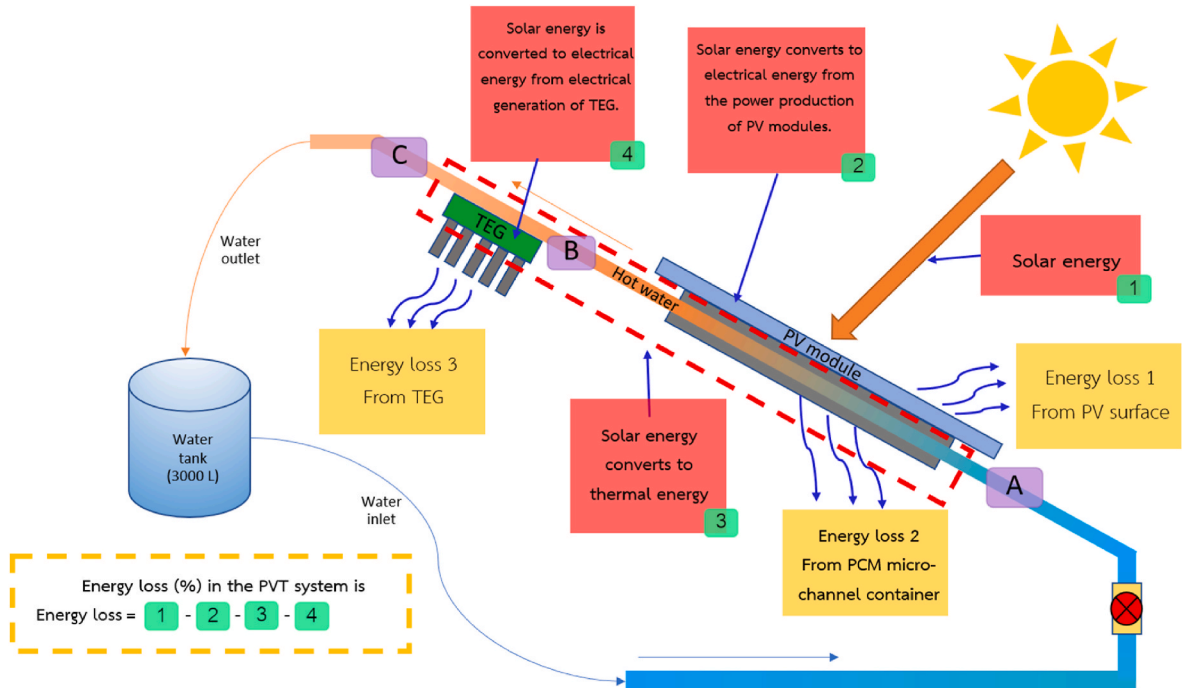


Fig. 5. Schematic of energy balance.

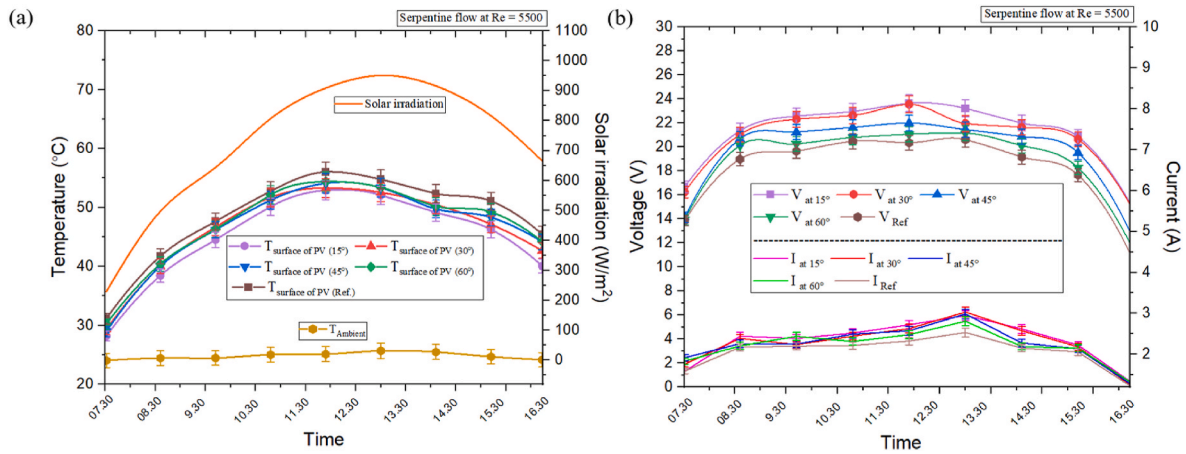


Fig. 6. (a) Surface temperature (b) Generation of voltage and current of PV modules for serpentine configurations varies at different inclination levels.

$$Ex_{out} = Ex_{PV} + Ex_{TEG} + Ex_{th} = P + P_{TEG} + \dot{Q}_U \quad (15)$$

$$Ex_{loss} = Ex_{in} - Ex_{out} \quad (16)$$

Where: Ex_{in} obtained through the sun is given by using the Petela-Landsberg-Press efficiency, T_{amb} is the ambient temperature in degree Celsius (°C), T_{sun} is the sun temperature ($T_{sun} = 5777 \text{ K} = 2500 \text{ °C}$), The exergy of PV (Ex_{PV}) and TEG (Ex_{TEG}) is equal the power output of PV, power output of TEG and the useful heat rate, respectively.

3. Results and discussions

To investigate the impact of different cooling methods on the overall efficiency of the PVT system, tests were carried out in November 2024. The experiments were conducted one day at a time for each inclination angle and were repeated daily until the data showed the comparable solar irradiation values for the entire day. Only then would the data be included in the data set for the average results. For example, the data for the 30° tilt angle is collected on November 5th, November 8th, and November 9th, 2024. November 6th and November 7th were not selected because they had significantly different solar irradiation curves when compared to the days that data collected. For each tilt angle, all experiments were conducted over a 12-day period of solar irradiation that was approximately $\pm 0.65\%$ of the selected experimental period. In addition, the installation location of the large water tank in section 2.4, which is inside a building with an insulated roof, ensures that the water inlet temperature remains within $\pm 0.14\%$ during each of the experiments. Experiments were conducted at the engineering faculty of Khon Kaen University, Thailand. The tested PV modules were installed on a flat area of a building at a height of about 12 m, in an open-air, shadowless location. Furthermore, the PVT system was tested under the environment of an average ambient temperature of 24.53 °C , an average wind speed of 2.2 m/s and an average humidity of 65% . The solar irradiation during the day was measured, as well as the ambient temperature, as shown as a top orange line for light intensity and a bottom dark yellow line for T_{amb} in Fig. 6(a). Therefore, this experiment controlled variables such as the temperature of the water inlet, ambient temperature, and solar irradiation.

3.1. Temperature reduction

Fig. 6(a) depicted the temporary changes in the temperature of the PV module at inclination angles of 15, 30, 45, and 60°. The results indicated that the PV temperature increased until peak time; after that, it decreased for all inclination angles of the PV module and reference module without cooling. The selection of the most appropriate phase change material for this climate hinged on the ambient temperature

(T_{amb}), which played a vital role in identifying the optimal PCM. When the melting temperature (T_{melt}) was between 3 °C plus the ambient temperature (T_{amb}) and 6 °C plus T_{amb} , the phase change material (PCM) took in less heat from the surroundings, which helped it use its melting energy more effectively. Consequently, PCM was expected to exhibit a melting temperature between 41 and 44 °C , while the MgO-PCM examined in this study demonstrated a melting point of 41.28 °C . In Fig. 6(a), the surface temperature of the reference module without installed PCM and TEG was higher than that of the PV with the PCM container at all inclination angles. The PCM effectively absorbed excess heat from the PV module during its phase change. As shown in Fig. 6(a), the surface temperatures of the PV module from 8:30 a.m. to 4:30 p.m. were greater than the PCM's melting temperature. As a result, it was confirmed that the PCM melted and changed phases during the experiment. Furthermore, Fig. 6(a) demonstrates that the temperature of the surface of the PV module with PCM cooling was lower than that of the PV module without cooling (reference PV module). However, inclination configurations for PV modules were more effective at reducing PV temperature than reference PV modules without cooling. Also, putting the micro-channel PCM container in PV modules with different tilt angles lowered the temperature of the PV surface. This was mainly due to the improved design of the structure inside the micro-channel PCM container, which enhanced heat transfer and reduced the temperature of the PV surface. Additionally, the current study clearly depicts the measured ambient temperature in Fig. 6(a). The average ambient temperature (outside temperature) is 24.53 °C , indicating a low level of heat. Furthermore, the water inlet temperature values range between 24.83 and 25.86 °C , indicating a setting that is near to the ambient temperature. Based on the analysis, it is clear that the PV module and the water outlet's increase in temperature are caused solely by heat transfer from the PV module, not by ambient temperatures. Fig. 6(b) illustrated the current and voltage of a PV module with and without the micro-channel PCM container at different inclination angles. It can be observed that all configuration types exhibited a peak period between 11:30 and 13:30 and showed a significant increase in the voltage (V) and current (I) values, followed by a gradual decrease until the day's end. The summary from Fig. 6(a)–(b) indicated that the inclination of the PV module at 15° , depicted by a violet line, corresponds to the minimum surface temperature, which subsequently resulted in the maximum values of voltage and current. The findings indicated that an angle of 15° also corresponds to the peak solar irradiation striking the surface. Optimal irradiation incidence on the surface of the PV panel was achieved when the azimuth angle closely matched the latitude angle. The latitude of the testing location played a crucial role in this phenomenon. The PV modules were positioned at an angle corresponding to the local latitude, approximately 16° , aligning with the latitude of Khon Kaen City, in accordance with the Earth's annual orbit around the sun. The

results aligned with the work of Maghrabie et al. [36].

3.2. Electrical analysis

Fig. 7(a)–(b) illustrated the power generation and electrical efficiency of PV modules in serpentine configurations. The findings showed that the power output rose for both the modified PV module and the reference PV module without cooling until peak time, after which it steadily declined. The maximum power output corresponded to the inclination of the PV module; the transition from low angle to high angle was illustrated by the violet line, red line, blue line, green line, and brown line (reference PV module without cooling), respectively. This was because the previously mentioned reduction in PV temperature implied that a lowered temperature will enhance power output. An increase in power output led to a direct improvement in electrical efficiency. It was seen that when comparing the PV module's average electrical efficiency at each tilt angle with the reference PV module without cooling's average electrical efficiency, the electrical efficiency increased by approximately 11–29% at angles of 60° to 15°, respectively. The newly implemented designed container demonstrated significant potential for lowering the surface temperature of the PV module. This was attributed to the container's internal micro-channel structure and MgO-PCM, both of which exhibited excellent heat transfer capabilities and are highly effective in absorbing heat from the PV module. Furthermore, the PV module positioned at 15° is an optimal option for increasing electrical efficiency because this angle was appropriate for gathering solar energy at this testing location.

3.3. Thermal analysis

Fig. 8 depicted the average daily temperature of the water inlet and outlet across various inclination levels from all experiments conducted at different angles over a span of 12 days. The findings indicated that the temperature of the water inlet remained remarkably consistent across all inclination angles. The water inlet temperature remained stable due to the inclusion of a substantial water tank (3000 L) within a building that featured an insulated roof. Furthermore, once the water leaves the system, it needs to cover a significant distance of approximately 35 m before it can return to the water tank. Consequently, this setup ensured that the water inlet temperature remained consistent, with a maximum and minimum difference of about 0.14% throughout all the experiments. The average temperatures of the water inlet were 25.15 °C, 25.22 °C, 25.16 °C, and 25.31 °C at angles of 15°, 30°, 45°, and 60°, respectively. The temperature of the water outlets rose as a result of the water absorbing excess heat from the PCM in the container, with the highest value occurring at a tilt angle of 15° (indicated by the red line).

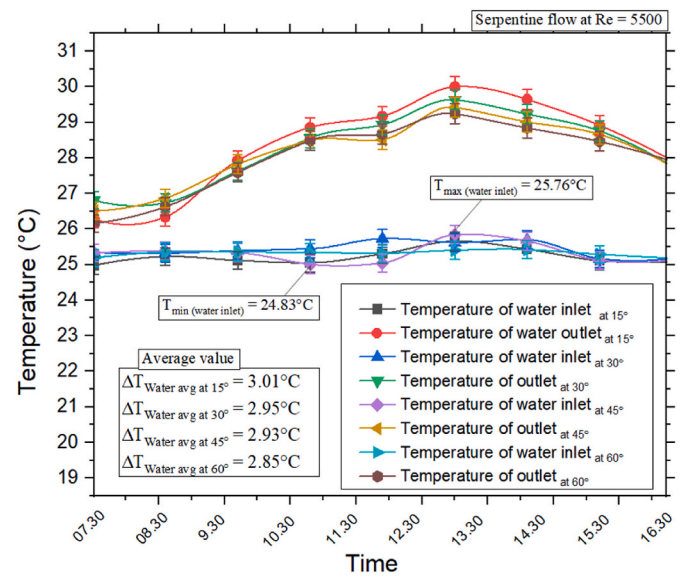


Fig. 8. Temperature of water inlet/outlet at different inclination levels.

Furthermore, the findings indicate that the average temperature difference between the water inlet and outlet was recorded at 3.01 °C, 2.95 °C, 2.93 °C, and 2.85 °C for angles of 15°, 30°, 45°, and 60°, respectively.

Fig. 9 indicated a slight variation in thermal efficiency. Different angles produced varying degrees of thermal efficiency, ranging from around 1% to 2%. The best thermal efficiency was reached at a tilt angle of 15°, giving an average value of 68.56%. This was because the PV's surface temperature was the coolest, as shown in Fig. 6. This means that the MgO-PCM in the container can absorb the maximum amount of heat at this tilt angle. Thermal efficiency was improved because the newly designed PCM micro-channels container can absorb more waste heat from the PV module; this led to the peak heat accumulation of water within the tube. Thus, when the temperature difference between the water inlet and outlet was high, thermal efficiency also increased. Furthermore, this hybrid active and passive PV cooling system with a novel design container can recover about 70% more energy when compared to the reference PV module without cooling.

Fig. 10 depicted the comparison between previous studies and the present study on overall efficiency. However, this study focused on comparing thermal efficiency. Various types of PVT systems were evaluated to determine their optimal performance. This effect was due

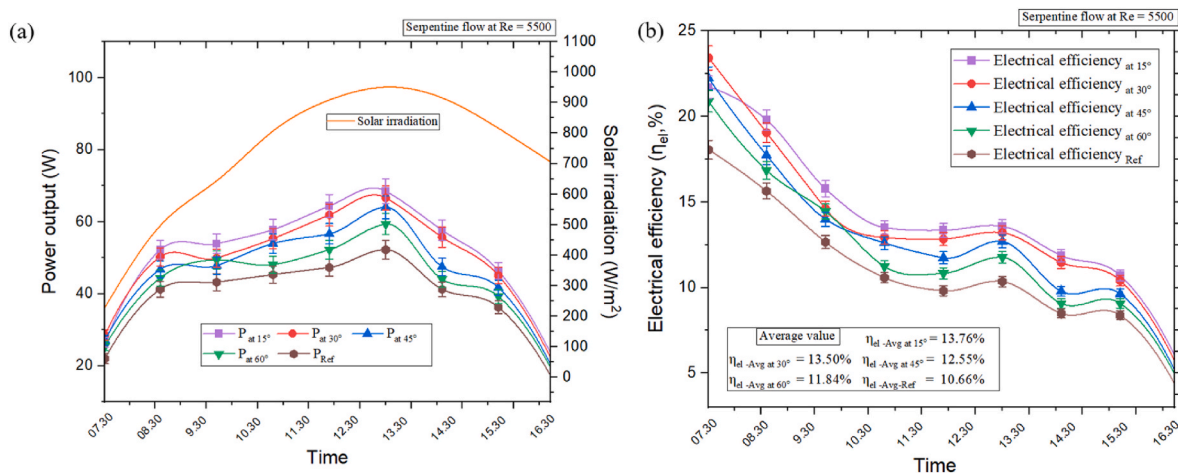


Fig. 7. (a) Power generation (b) Electrical efficiency of PV modules for serpentine configurations varies at different inclination levels.

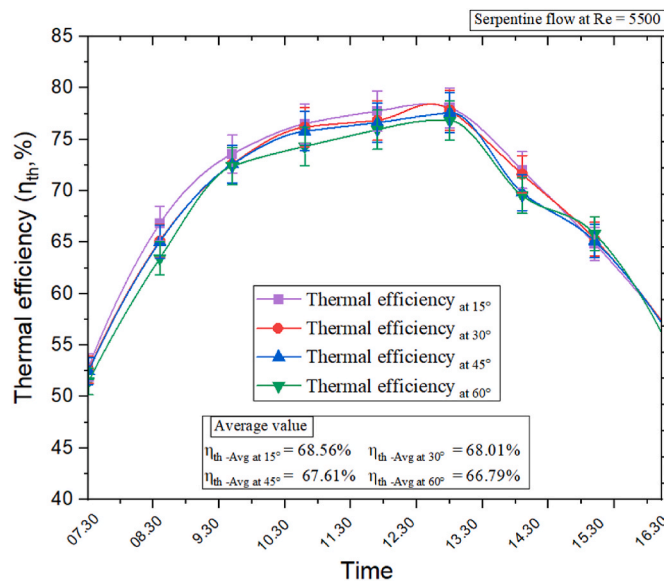


Fig. 9. Thermal efficiency of PV modules for serpentine configurations varies at different inclination levels.

to the fact that heat from solar irradiation accumulated significantly on the PV surface. All studies used a PVT system that measured overall efficiency, making it suitable for comparison. Table 5 presented the best conditions for each of the previous studies. Based on Fig. 10 and Table 5, the thermal efficiency of all studies was more influential to overall efficiency than electrical efficiency, except for the work of Agrawal et al. [64], which had the lowest overall efficiency compared to other studies. The accumulation of heat from solar energy on the surface of the PV results in a notable difference in temperature between the inlet and outlet. In addition, the flow of water within the tube can assist in the absorption of high temperatures (excess heat) from the PV module, leading to a variation between the inlet and outlet water temperatures and thereby enhancing thermal efficiency. Furthermore, the micro-channel PCM container designed in this study contributes to higher overall efficiency than previous research. Thermal energy can be recovered by up to 68.56%, resulting in the highest overall efficiency in

the serpentine flow designed with a 15° tilt angle and $Re = 5500$.

3.4. Thermoelectric generators analysis

The aim of this study was to create a PVT-TEG power generation system that improved the electrical efficiency of PVT panels by turning the heat from the water tube in the PCM micro-channel container into electricity for TEGs. Fig. 11(a) demonstrated the power output of TEG, and Fig. 11(b) showed the electrical efficiency of TEG. In addition, Fig. 11(c) indicated temperatures of TEG. It was worth noting that at all four tilt angles of the PV module, the temperature variation curves of the TEG almost overlapped. The power output, electrical efficiency, and cold/hot temperatures of the TEG exhibited similar results. Although the water entered the TEG at a tilt angle of 15, which corresponded to a higher temperature than other tilt angles, the hot and cold sides of the TEG exhibited slight variations across all cases. Consequently, the specific power of the TEG did not indicate significant differences, as illustrated in Fig. 11(d). Nevertheless, the temperature difference between the water inlet and exhaust influenced both the electrical efficiency of TEG and the power output. The maximum average power output value and electrical efficiency value of TEG were determined to be 0.61 W and 0.169%, respectively, at 15°, as indicated by the results. Fig. 11(d) presented that a tilt angle of 15° resulted in the maximum TEG specific power at 0.000075 W/(°C mm²). Furthermore, the TEG specific power can be understood as its power generation efficiency, as this index value showed power output in relation to differences in temperature and area. However, in the context of the PVT system, the TEG's performance was compared to the solar irradiation it got, which is known as electrical conversion of TEG (electrical efficiency of TEG, η_{TEG}).

3.5. Overall efficiency

Figs. 12 and 13 data revealed that when inclination levels grew, electrical and thermal efficiency values declined, whereas TEG efficiency values showed very little change. However, thermal efficiency was the most important factor influencing total efficiency; hence, overall efficiency will increase or decrease depending on thermal efficiency. Fig. 12 presented that the greatest average overall efficiency was 82.49% at the 15° inclination angle level, while the minimum average overall efficiency was 78.79% at the 60° inclination angle level.

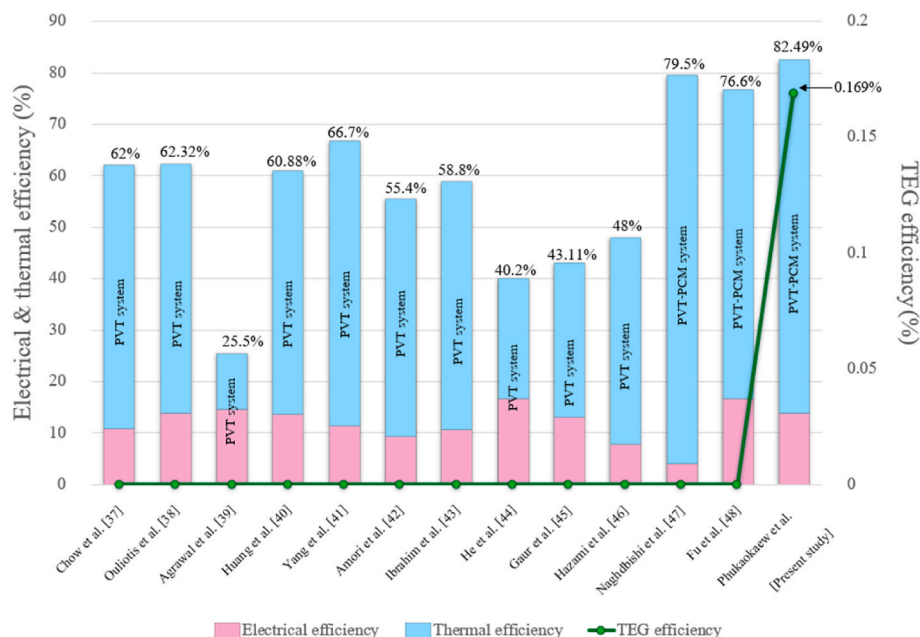


Fig. 10. Comparison of efficiency's present study to several previous studies on related PVT systems.

Table 5

Comparison of efficiency's present study to several previous studies on related PVT systems.

Author's Name [Reference]	Methods and benefits	PVT system	PCM	Overall efficiency
Chow et al. [37]	The monocrystalline glazed module with the flat-box PV/T collector and 175-liter water storage tank was installed. The system was tested under environmental conditions where the average solar irradiation level was 711 W/m ² at a tilt angle of 20°, and the mass of water in the thermosyphon system relative to the collector surface area was measured at 96.6 kg/m ² .	✓		62%
Ouliotis et al. [38]	The setup for the study included configurations of PVT glazed and unglazed systems, as well as PVT systems that were fitted with diffuse reflectors. A thermosiphonic water circulation system, which included storage, connected all these systems together. The parameters were defined in the following manner: The water recorded a temperature of 42 °C, whereas the air was measured at 40 °C.	✓		62.32%
Agrawal et al. [39]	The glazed hybrid micro-channel photovoltaic thermal (MCPVT) was evaluated for performance; the micro-channel has been placed between Tedlar and insulation. The experiment was performed in indoor conditions with a mass flow rate of 0.000108 kg/s, necessitating air circulation and monocrystalline enamel.	✓		25.5%
Huang et al. [40]	The unglazed flat-plate polycrystalline PVT module was equipped with an integrated pump control system. The PV/T collector, which was manufactured with copper tube and copper sheet, was installed on the back of the PV module to absorb waste heat. The system was tested under average solar irradiation of 510 W/m ² , an ambient air temperature of 25.32 °C, and a water temperature of 11.65 °C.	✓		60.88%
Yang et al. [41]	The single crystalline PVT with functionally graded material (FGM) and water tube was installed at the back side of the PV to reduce temperature. The PVT was evaluated under an average solar irradiance of 1100 W/m ² , a water inlet temperature of 20 °C, and a mass flow rate of 66 ml/min.	✓		66.7%
Amori et al. [42]	The flat plate PVT polycrystalline module was cooled with air. PV had a	✓		55.4%

Table 5 (continued)

Author's Name [Reference]	Methods and benefits	PVT system	PCM	Overall efficiency
	temperature of 59.55 °C in the absence of a cooling system. The dual-pass PVT cooling system reduced the temperature by 15.52 °C, bringing it down to 44.3 °C using a single duct and an airflow rate of 0.0991 kg/s.			
Ibrahim et al. [43]	The building-integrated photovoltaic thermal (BIPVT) system included multi-crystal PV and a spiral flow absorber for BIPV. The spiral flow absorber was made of rectangular hollow tubes of stainless steel material with dimensions of 12.7 × 12.7 mm, and the spiral coil has water inlet and outlet. The system was conducted with a mass flow rate of 0.027 kg/s, an average ambient temperature of 33 °C, an average water outlet of 41 °C, and an average solar irradiation of 690 W/m ² .	✓		58.8%
He et al. [44]	The heat pipe photovoltaic/thermal (PV/T) with a 150-liter water storage tank was installed. The system was tested in winter with an average solar irradiation of 500 W/m ² and an average water inlet temperature of 21 °C.	✓		40.2%
Gaur et al. [45]	The building-integrated semitransparent photovoltaic/thermal (BiSPVT) system had a combined solar panel and heating system that used a flat plate monocrystalline design with forced air circulation, and it was compared to systems with and without ducts. It was found that the system without ducts gave the best efficiency, which was tested with the condition of an inclined angle of 34° from the horizontal, a mass flow rate of 0.85 kg/s, an average solar irradiation of 422 W/m ² , and an average ambient temperature of 26.2 °C.	✓		43.11%
Hazami et al. [46]	The objective of the investigation was to determine the potential benefits of investing in a cutting-edge solar composite system (SCS) that includes a PVT solar collector to meet the demand for space heating. The following parameters were examined in this study: an average ambient temperature of 26 °C, an average wind speed of 1.5 m/s, a mass flow rate of 0.027 kg/s, and an average solar irradiance of 584 W/m ² .	✓		48%

(continued on next page)

Table 5 (continued)

Author's Name [Reference]	Methods and benefits	PVT system	PCM	Overall efficiency
Naghdishi et al. [47]	A hybrid PVT/PCM system had been created to test how water/glycol-based nanofluids worked as active cooling agents, along with organic paraffin wax (PCM) that melted at 57–60 °C and acted as the passive cooling medium. The system was tested with an average solar irradiance of 792 W/m ² , a mass flow rate of 30 kg/h, and an average ambient temperature of 32 °C. Additionally, the maximum reduction in temperature of the PV module was observed to be 21%.	✓	✓	79.5%
Fu et al. [48]	An experimental and numerical investigation was conducted on the performance of a multilayer structured PVT system incorporating PCM, specifically paraffin wax with a melting point of 32 °C (RT35). The system employed a copper tube positioned behind the module for cooling, utilizing conditioned water with a flow rate of 750 L/h, an average solar irradiance of 598 W/m ² , and an average ambient temperature of 15 °C.	✓	✓	76.6%
Phuakaew et al. [Present study]	The PVT system used MgO-PCM within the micro-channel container and varied angles of tilt to optimize overall efficiency. The MgO-PCM was a combination of lauric acid PCM, which had a melting point of 42.26 °C, and the MgO nanoparticle. The best case occurred in serpentine configuration at Re = 5500 and a tilt angle of 15°, an average ambient temperature of 24.53 °C, an average wind speed of 2.2 m/s, and average solar irradiation of 815.11 W/m ² .	✓	✓	82.49%

Meanwhile, the PV reference without cooling had an average overall efficiency of 10.66% due to the absence of TEG equipment and PCM micro-channel containers. Fig. 13 illustrated the enhanced clarity of bar graphs in presenting electrical, thermal, and TEG efficiency values. The blue bar graphs represented thermal efficiency values, the yellow bar graphs represented electrical efficiency values, and the red bar graphs represented TEG efficiency values. From the bar graph data, it can be observed that the blue bar graph, electrical efficiency, was relatively smaller than thermal efficiency. Based on the results, thermal efficiency had the greatest impact on overall efficiency, followed by PV electrical efficiency and TEG, which had the least effect. In addition, the experimental angle of 15° yielded the highest overall efficiency compared to the experiments at other angles. In the present investigation, a combination of various cooling techniques was employed, including active cooling through the use of a water pump and passive cooling by integrating PCM with nanoparticles. This system incorporated two distinct

cooling methods; it can be called a hybrid active and passive PV cooling system. In addition, the contributions of PVT, PCM, and TEG technologies are as follows: The PVT system plays a crucial role in the thermal term. The water inlet temperature values around 25 °C and the water outlet temperature around 28 °C demonstrate this effect. It means heat is absorbed from the PV module and transferred to the water. This observation allows for thermal efficiency analysis. Additionally, the PCM technologies can be observed in the temperature graph, which shows the decrease in the surface temperature of the PV module when the PCM container is installed, resulting in increased electrical efficiency. Furthermore, TEG, an add-on installation, generates additional electricity by allowing hot water to flow to the hot side of the devices. Thus, three technologies contribute to the overall efficiency of the PVT system, defined as the conversion of solar energy into useful energy, as depicted in Figs. 12 and 13.

3.6. Energy analysis

Fig. 14 depicted the energy pathway in the system. First, the PV module collected solar energy from the sun, which was assumed to be 100%. Solar energy was converted to electrical energy at the PV module at a rate of 10–14% at angles ranging from 60° to 15°, with some energy lost in the process. Subsequently, the PCM micro-channel container, which was located behind the PV module, absorbed the PV module's waste heat and transferred it to water in the tube. Hot water ran through the TEG equipment, producing around 0.169% of electrical energy. In the two processes outlined above, solar energy transformed to thermal energy at various inclination levels ranging from 60° to 15° at a rate of around 66–68%. The PCM micro-channel container and TEG equipment also experienced energy loss. The PVT system lost approximately 17.83 to 23.83% of its energy.

Fig. 15(a) illustrated the trends in thermal efficiency and electrical efficiency, where the red line denoted thermal efficiency, and the blue line indicated electrical efficiency. It was found that as the Reynolds numbers (Re) increased, there was a corresponding slight increase in electrical efficiency. In a similar manner, as the Reynolds numbers (Re) increased, thermal efficiency improved until reaching Re = 5500, which represented the peak value of thermal efficiency, after which it began to decline gradually. In the PVT system, thermal efficiency significantly influenced overall efficiency more than electrical efficiency; therefore, the experimental condition for this study was Re = 5500. Fig. 15(b) illustrated the trends in thermal efficiency alongside the water output temperature, where the green line indicated the temperature of the water output. As the Reynolds numbers increased, it became evident that the temperature's water output decreased, given that the Reynolds number was associated with the water flow rate. The findings suggested that at elevated Reynolds numbers, water exhibited increased flow rates, leading to a reduced outlet water temperature. The thermal efficiency was still good, though, because high Reynolds numbers caused a lot of mass to flow, and according to equation (4), the mass flow rate value was used for multiplication. Consequently, it was recommended that if hot water (highest temperature) was required, a low Re level should be selected, even though it may also yield low thermal efficiency. However, if the highest thermal efficiency was desired, Re = 5500 was the most suitable value. In a similar manner, Fig. 15(c) displayed the trends in electrical efficiency in contrast to the water output temperature, which exhibited opposing directions. The optimum system was determined by user preferences: when high-temperature hot water was needed, a low Reynolds number was more appropriate. On the other hand, when there was a demand for increased electricity production, a higher Re level proved to be more appropriate. However, this research selected the case Re = 5500 because it offered the highest overall efficiency.

3.7. Exergy analysis

The second law of thermodynamics was employed to conduct the

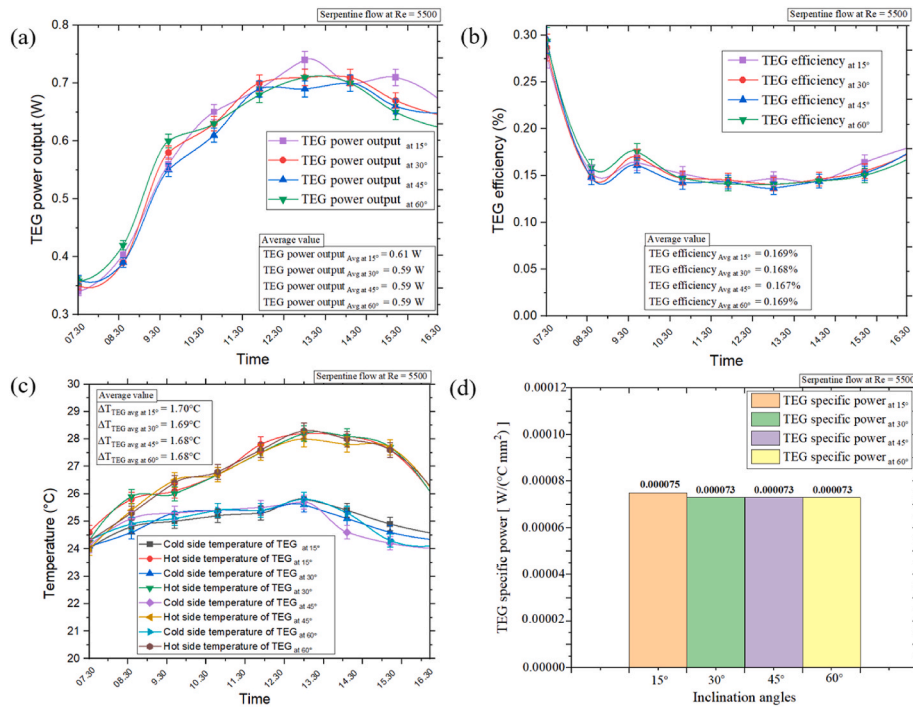


Fig. 11. (a) Power generation (b) Electrical efficiency (c) Temperatures of TEGs (d) TEG specific power for serpentine configurations vary at different inclination levels.

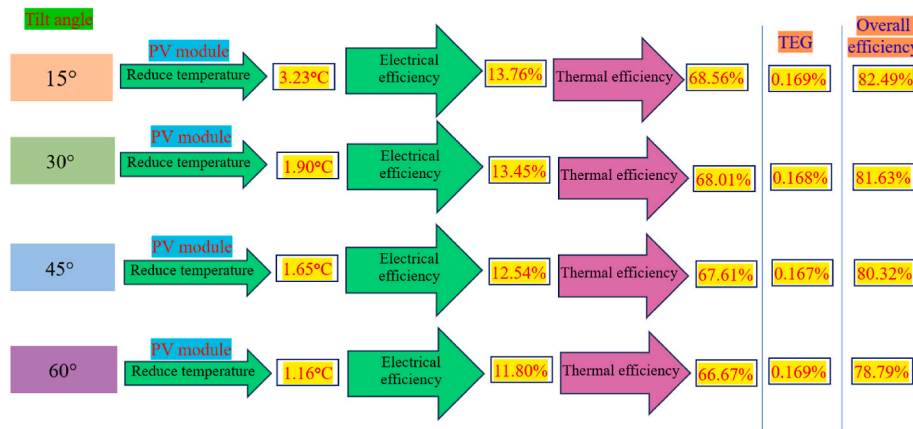


Fig. 12. Comparison between each tilt angle of all results.

exergy analysis, which evaluated the utmost work that can be extracted from the PV/T system in relation to its environmental context. The difference between energy efficiency and exergy efficiency was as follows:

Energy efficiency: Energy efficiency quantifies the useful energy output derived from a specific quantity of energy input. It was generally represented as a percentage. It emphasized the transformation of energy from one form to another, without considering its quality.

Exergy efficiency: Exergy efficiency assessed the effectiveness of a system in transforming available energy (exergy) into useful work or energy. Exergy represented the maximum useful work that can be extracted from a system in a specific state. This considers both the quantity and quality of energy. Exergy takes into account the capacity for work and the irreversibility presented in the process.

Exergy efficiency (also known as the second law efficiency or rational efficiency) variation plotted of the enhanced PVT-TEG power generation system was depicted in Fig. 16 (a). It was found that the

exergy efficiency of PVT-TEG panels was higher than that of reference PV without cooling in tests that were done at different angles for each trial. The PVT panel's exergy efficiency also went up: 12.78% at an angle of 15°, 12.35% at an angle of 30°, 11.53% at an angle of 45°, 10.89% at an angle of 60°, and 9.12% for the PV reference without cooling. These findings were consistent with those of Azeez et al. [49] and Selimefendigil et al. [50]. Even though all tilt angles of testing produced comparable results, it was found that at 15°, the PVT system's exergy efficiency value was the highest for the same reason: the location of testing. The tilt angle of 15° was the most effective at converting available energy (exergy) into productive work or energy. Additionally, equation (15) presented that the exergy output included the power from the PVT system (P), the power from the TEG (P_{TEG}), and the useable heat rate ($\dot{Q}_U$). Therefore, it was apparent that at this angle, the sum reached its maximum intensity, which results in the highest possible exergy efficiency. This was the case because the sun reached its peak energy at this tilt angle. An increase of 40.13% was found at an angle of 15° when

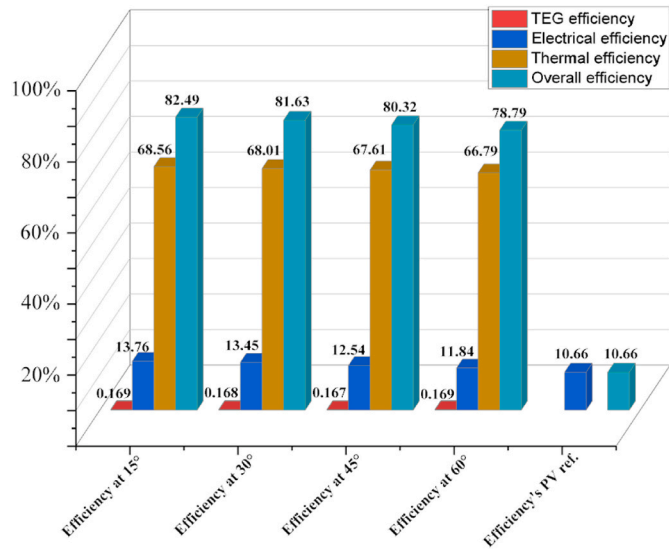


Fig. 13. Average electrical efficiency, TEG efficiency, thermal efficiency, and overall efficiency at different levels of inclination.

compared to the exergy value for the PV reference without cooling.

Fig. 16 (b) depicted exergy loss of the PVT system and PV reference without cooling in units of watts. Based on results, it was found that the exergy loss of PVT-TEG systems was significantly lower than that of reference PV without cooling. This finding aligned with the previously mentioned exergy efficiency. At tilt angles of 15°, 30°, 45°, and 60°, the exergy loss was equal to 359.76 W, 361.47 W, 364.8 W, and 367.4 W, respectively, while it equals 371.9 W for reference PV without cooling. The exergy loss in each scenario was assessed through exergy efficiency, which diminished as the capacity for exergy conversion rose. This progressively diminished as the angle of the tilt escalated in succession. Thus, at an angle of 15°, the highest exergy value corresponded to the lowest exergy loss. In comparison to reference PV without cooling, this measurement resulted in an approximate reduction of exergy loss by 2%.

4. Conclusion

The current research utilizes a hybrid method by incorporating active cooling via a water pump and passive cooling through the integration of PCM with nanoparticles. This system is called photovoltaic thermal (PVT), which produces electricity and hot water. The tests are conducted at Khon Kaen University's Faculty of Engineering to determine how different cooling methods affect the thermal and electrical performance of PV modules installed at different angles with their faces facing south. The experimental setup utilizes inclination angles of 15°,

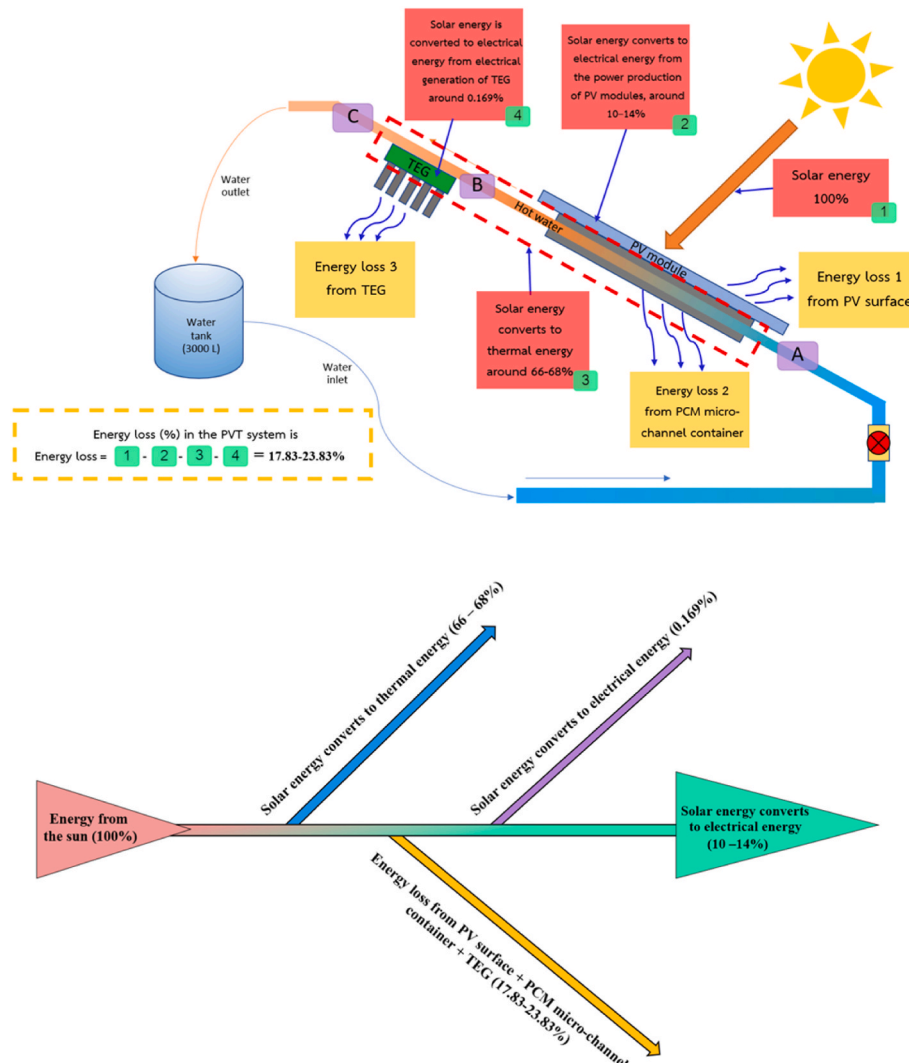


Fig. 14. Energy balance and fishbone diagram of system.

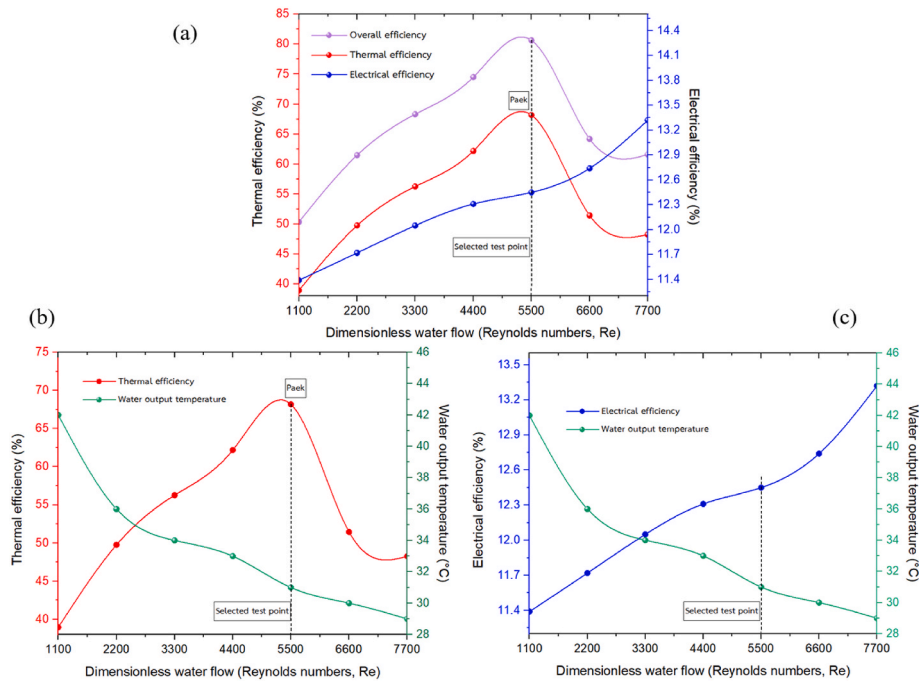


Fig. 15. Trend of thermal efficiency, electrical efficiency, and water output temperature.

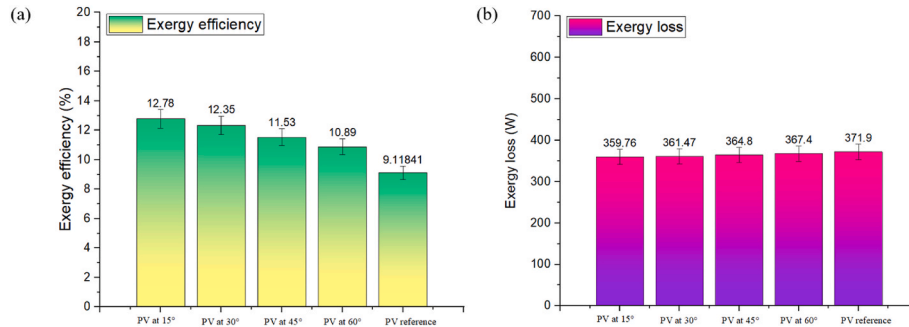


Fig. 16. (a) Exergy efficiency of PVT system and PV reference (b) Exergy loss of PVT system and PV reference.

30°, 45°, and 60° compared to reference PV modules without cooling. The system also includes a TEG installation. The results suggest that enhanced photovoltaic modules' performance to generate electricity is influenced by the thermal conductivity of the MgO-PCM and micro-channel containers. The novel design of the PCM container, which features a micro-channel structure, contributes to the reduction of the PV surface. The investigation of inclination angles demonstrated that the tilt angle of 15° has a maximally lower PV surface temperature, by 3.23 °C (6.76%), thereby improving electrical efficiency by 29.08%. This result also corresponds to the experimental location's latitude of 16°. In addition, the maximum water inlet/outlet temperature difference (ΔT_{water}) is 2.51 °C at the tilt angle of 15°, resulting in the maximum thermal efficiency of the PVT being 68.56% and the lowest energy loss being 17.83%.

Three TEGs are inserted before the hot water is drained at the water tube outlet, rather than at the back of the PV module in the other works. Reusing the high-temperature water from cooling the system to generate power through the TEG increases the system's overall efficiency. The maximum average power output value and electrical conversion of TEG are determined to be 0.61 W and 0.169% at the tilt angle of 15°. Furthermore, exergy efficiency measures the amount of exergy lost in each condition; it decreases as the amount of exergy available for conversion increases. This exergy value gradually decreases as the tilt angle

increases. The 15° tilt angle produces the highest exergy value of 12.78% and also has the lowest exergy losses of 359.76 W.

The study's limitations are the variations in solar irradiation that result from the outdoor testing conditions that occur every day. To address this, the experiments were conducted one day at a time for each tilt angle and repeated over a period of several days. Subsequently, three days with comparable solar irradiation values throughout the day were selected for analysis. The aim was to ensure that the experimental conditions were comparable. Another limitation involves the non-uniform distribution of nanoparticles within PCM when used in a conventional PVT system. Due to the test, some of the PCM underwent a change from solid to liquid as they absorbed heat from the PV module, which led to their melting. Additionally, the tilt of the PV module might cause the liquid form of PCM to move to the bottom of the container. This could result in a non-uniform, uneven distribution of nanoparticles within the PCM. Future work will focus on exploring alternative system designs to achieve more consistent nanoparticle dispersion and assess long-term performance. Furthermore, restrictions for outdoor investigations include unexpected rain. To accomplish this, plan the experiment, verify the weather forecast, and provide rain protection for the experimental equipment.

CRediT authorship contribution statement

W. Phuakaew: Writing – original draft, Investigation, Formal analysis, Data curation. **A. Suksri:** Validation, Methodology, Formal analysis. **A. Intaniwet:** Validation, Formal analysis. **K. Punyawudho:** Validation, Formal analysis. **T. Paphapote:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **T. Wongwuttanasatian:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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